

Appium - Installation & Setup

A complete guide to prepare the environment, install, configure, validate and perform the first functional use of the tool.

Manual objective

This manual guides you through the complete installation and setup process, from preparing the environment to final validation. Follow the steps in order and use the checks throughout the guide to confirm that each configuration was completed correctly.

Official references

<https://appium.io/>

<https://appium.io/docs/en/latest/>

<https://appium.io/docs/en/latest/quickstart/install/>

Drivers

Appium requires drivers. UiAutomator2 targets Android; XCUITest targets iOS/iPadOS on macOS.

How to start safely

Start from the official Appium site and documentation. Confirm prerequisites, supported versions, operating system and required permissions. Learn in a controlled environment before using it in a critical workflow.

First QA-oriented use

Pick one small, verifiable task in Appium, configure only what is necessary, run a simple example and confirm the outcome. Repeat with different data or conditions before adding abstractions.

Common issues and troubleshooting

When Appium fails, verify versions, dependencies, permissions, connectivity, data and environment before assuming a product defect. Preserve the original error and reduce the issue to the smallest reproducible case.

Installation — dependency chain

Appium is an extensible server. For Android prepare Node.js/npm, Appium, Android Studio/SDK, Java when required by the driver, an emulator/device and UiAutomator2.

Install Appium

With Node/npm working, install Appium according to official docs and validate `appium --version`. Start appium to confirm the server launches.

Android driver

Run `appium driver install uiautomator2`, then `appium driver doctor uiautomator2` to identify missing required/optional dependencies.

Android SDK and variables

Validate `adb --version` and `adb devices`. If `adb` is missing, review SDK location and add `platform-tools` to `Path`; configure recommended environment variables when needed.

First session

Start an emulator or connect an authorized USB-debug device, confirm `adb devices` and create an Appium session with compatible capabilities.

Troubleshooting

Common issues include Node/npm `PATH`, missing SDK/`adb`, incompatible JDK/`JAVA_HOME`, missing driver or unauthorized device. Use Doctor before blaming the application.